KIAA1549 fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes KIAA1549 gene fusions from the msk_impact_50k_2026 study, covering 44 fusion events. 40/41 fusions with known frame status (97.6%) were in-frame. 22/44 fusions (50.0%) retained the UPF0606 protein KIAA1549 alpha/beta domain 2 domain. Domain retention was tested with Fisher's exact test (p=0.927235, not statistically significant at alpha=0.05). No literature benchmark is configured for KIAA1549.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 44 analyzed fusion events, identifying 2 distinct fusion partner genes. The most frequent partner was BRAF, observed in 43 events.

Domain retention

UPF0606 protein KIAA1549 alpha/beta domain 2 domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 19/40 (47.5%) of in-frame fusions retained the domain, p=0.927235 (not statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 9.9999e-06.

Domain disruption

Domain-disruption analysis was skipped: KIAA1549 has no disruption_required_domains configured; domain-disruption analysis was skipped.

Junction concentration

54% of 39 fusions with this gene as the 5' partner join at exon 16, against 6% expected if breakpoints fell at random along the gene; this is the most over-represented junction (empirical p=9.9999e-06, corrected for testing every exon, statistically significant at alpha=0.05). The random-breakage null assumes a breakpoint anywhere in the gene would be detected. Capture panels such as MSK-IMPACT tile only selected introns, so concentration on panel data is partly by assay design and this p-value overstates the evidence for biological concentration. Confirm in a cohort sequenced without intron selection (whole genome or RNA).

Cutpoint detection

Cutpoint detection scanned 39 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 1687 aa (exon 16; label-permutation p=0.000999001). This boundary is descriptive: retained/lost status follows from breakpoint position, so a clean split exists by construction and the p-value is not evidence of selection. This is -15 aa from the nearest configured domain boundary at 1702 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" (n=21) and "not_retained" (n=18). For Frame_status, retained: 18/19 (94.7%, 95% CI 75.4%-99.1%); not_retained: 17/17 (100.0%, 95% CI 81.6%-100.0%). Welch's t-test comparing Tumor_variant_count between groups gave means of 32.8571 vs 43.6667, p=0.498149 (not statistically significant at alpha=0.05).

Composite evidence score

Composite evidence score reported: domain_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption_required_domains not configured for this gene).

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Expression association

Expression association reported: No mRNA expression data was available for KIAA1549 in this cohort; expression-association analysis was skipped.

Genomic position recurrence

Genomic position recurrence produced results for this run (17 row(s) in genomic_bin_recurrence, 0 row(s) in exact_position_recurrence, 3 row(s) in protein_position_genomic_spread); see the accompanying table.

Joint partner

Joint partner reported: KIAA1549 has no gene_pair configured; joint-partner analysis was skipped.

Mechanistic interpretation

Mechanistic interpretation produced results for this run; see the accompanying data for details.

Mutation cooccurrence

Mutation cooccurrence reported: KIAA1549 has no mutual_exclusivity_targets configured; co-occurrence/mutual-exclusivity analysis was skipped.

Window detection

Window detection produced results for this run (34 row(s) in window_scan, 12 row(s) in top_windows); see the accompanying table.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
BRAF	43	0.9772727272727273	0.5000004342923104	None	0.0	0.7893232736922304	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.6213928176551484	1
STK11	1	0.022727272727272728	0.5000004342923104	None	0.23132137030995115	0.7893232736922304	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.30681102913371894	2

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
1344	0	21	3	15	0.0	0.08928766823503669	1.0492085185727995
1449	0	21	4	14	0.0	0.03720319509793195	1.4294197602844054
1643	0	21	17	1	0.0	3.527950929462398e-10	9.452477464133164
1687	0	21	18	0	0.0	1.6036140588465443e-11	10.79490014495537
1730	1	20	18	0	0.0	3.0468667118084345e-10	9.516146544002542
1749	20	1	18	0	0.0	1.0	-0.0

domain_retention tables

frame_domain_contingency_table

18	3
17	1

frame_domain_contingency_tables_by_tumor_type

stratum	table
ACRM	[0, 0], [1, 0]
AGNG	[1, 0], [0, 0]
BRCNOS	[0, 0], [1, 0]

stratum	table
CUP	[0, 1], [0, 0]
ERMS	[0, 0], [1, 0]
EVN	[0, 0], [1, 0]
GB	[1, 0], [4, 0]
GNG	[1, 0], [0, 0]
ILC	[0, 0], [1, 0]
MNET	[0, 0], [2, 0]
MPNST	[1, 0], [0, 0]
NST	[1, 0], [0, 0]
PAST	[13, 1], [6, 1]
PMA	[0, 1], [0, 0]

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_key_domain	UPF0606 protein KIAA1549 alpha/beta domain 2	39	21	18	0	0.8246007984031937

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
4	39	39	1.0

frequency tables

Partner_gene_counts

Partner_gene	Event_count	Raw_event_count	Sample_count	Patient_count
BRAF	43	43	43	38
STK11	1	1	1	1

genomic_position_recurrence tables

genomic_bin_recurrence

chromosome	bin_start	bin_end	n_events	event_ids
7	138537000	138538000	4	EVT-P-0014191-T01-IM6-5, EVT-P-0020125-T01-IM6-12, EVT-P-0032982-T01-IM6-19, EVT-P-0056952-T01-IM6-34
7	138539000	138540000	4	EVT-P-0013777-T01-IM5-4, EVT-P-0022404-T01-IM6-14, EVT-P-0038818-T01-IM6-44, EVT-P-0043874-T01-IM6-27
7	138546000	138547000	4	EVT-P-0015540-T01-IM6-7, EVT-P-0017693-T01-IM6-9, EVT-P-0062963-T01-IM7-37, EVT-P-0069024-T04-IM7-42
7	138548000	138549000	4	EVT-P-0014381-T01-IM6-6, EVT-P-0029361-T01-IM6-18, EVT-P-0035153-T01-IM6-43, EVT-P-0052411-T03-IM6-31
7	138538000	138539000	3	EVT-P-0004032-T01-IM5-20, EVT-P-0062070-T01-IM7-36, EVT-P-0066030-T01-IM7-41
7	138545000	138546000	3	EVT-P-0027592-T01-IM6-15, EVT-P-0045683-T02-IM6-29, EVT-P-0049925-T01-IM6-30
7	138550000	138551000	3	EVT-P-0005121-T02-IM5-21, EVT-P-0040760-T01-IM6-25, EVT-P-0043372-T01-IM6-26
7	138541000	138542000	2	EVT-P-0028611-T01-IM6-16, EVT-P-0066022-T01-IM7-40
7	138542000	138543000	2	EVT-P-0009097-T01-IM5-2, EVT-P-0064310-T04-IM7-38
7	138544000	138545000	2	EVT-P-0044270-T02-IM6-28, EVT-P-0058584-T01-IM7-35
7	138551000	138552000	2	EVT-P-0012575-T01-IM5-3, EVT-P-0055099-T03-IM6-33
7	138543000	138544000	1	EVT-P-0019127-T01-IM6-11
7	138549000	138550000	1	EVT-P-0016557-T01-IM6-8
7	138556000	138557000	1	EVT-P-0034117-T01-IM6-1
7	138567000	138568000	1	EVT-P-0035844-T03-IM7-23
7	138574000	138575000	1	EVT-P-0064872-T01-IM7-39
7	138575000	138576000	1	EVT-P-0021127-T01-IM6-13

protein_position_genomic_spread

protein_position_aa	n_events	n_distinct_genomic_positions	genomic_span_bp	spans_multiple_chromosomes
1749	19	19	8465	False
1643	13	13	5699	False
1344	3	3	7560	False

junction_concentration tables

junction_counts

role	junction_exon	count	role_event_count	observed_share	random_breakage_share	excess_neg_log10_p
five_prime	16	21	39	0.5384615384615384	0.06388956500573734	14.782373303246747
five_prime	15	13	39	0.3333333333333333	0.04639938801766404	7.92141894655934
five_prime	10	3	39	0.07692307692307693	0.09000312945512709	0.15856605973270665
five_prime	12	1	39	0.02564102564102564	0.05765847213046351	0.04510957914559799
five_prime	17	1	39	0.02564102564102564	0.05421607149066379	0.05243526345697713

window_detection tables

window_scan

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
1249	1449	200	4	35	0	21	0.0	0.03720319509793195	1.4294197602844054	EVT-P-0021127-T01-IM6-13, EVT-P-0034117-T01-IM6-1, EVT-P-0035844-T03-IM7-23, EVT-P-0064872-T01-IM7-39
1344	1544	200	4	35	0	21	0.0	0.03720319509793195	1.4294197602844054	EVT-P-0021127-T01-IM6-13, EVT-P-0034117-T01-IM6-1, EVT-P-0035844-T03-IM7-23, EVT-P-0064872-T01-IM7-39
1443	1643	200	14	25	0	21	0.0	2.028571784440879e-07	6.692809619443533	EVT-P-0005121-T02-IM5-21, EVT-P-0012575-T01-IM5-3, EVT-P-0014381-T01-IM6-6, EVT-P-0015540-T01-IM6-7, EVT-P-0016557-T01-IM6-8, ... (14 total)
1449	1649	200	14	25	0	21	0.0	2.028571784440879e-07	6.692809619443533	EVT-P-0005121-T02-IM5-21, EVT-P-0012575-T01-IM5-3, EVT-P-0014381-T01-IM6-6, EVT-P-0015540-T01-IM6-7, EVT-P-0016557-T01-IM6-8, ... (14 total)
1487	1687	200	14	25	0	21	0.0	2.028571784440879e-07	6.692809619443533	EVT-P-0005121-T02-IM5-21, EVT-P-0012575-T01-IM5-3, EVT-P-0014381-T01-IM6-6, EVT-P-0015540-T01-IM6-7, EVT-P-0016557-T01-IM6-8, ... (14 total)
1530	1730	200	15	24	1	20	0.014285714285714285	4.746857975591655e-06	5.32359376203339	EVT-P-0005121-T02-IM5-21, EVT-P-0012575-T01-IM5-3, EVT-P-0014381-T01-IM6-6, EVT-P-0015540-T01-IM6-7, EVT-P-0016557-T01-IM6-8, ... (15 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
1543	1643	100	13	26	0	21	0.0	1.054857327909257e-06	5.976806275808734	EVT-P-0005121-T02-IM5-21, EVT-P-0012575-T01-IM5-3, EVT-P-0014381-T01-IM6-6, EVT-P-0015540-T01-IM6-7, EVT-P-0016557-T01-IM6-8, ... (13 total)
1549	1749	200	34	5	20	1	5.714285714285714	0.16183389867600395	0.7909305033297682	EVT-P-0004032-T01-IM5-20, EVT-P-0005121-T02-IM5-21, EVT-P-0009097-T01-IM5-2, EVT-P-0012575-T01-IM5-3, EVT-P-0013777-T01-IM5-4, ... (34 total)
1554	1754	200	35	4	21	0	None	0.03720319509793194	1.4294197602844056	EVT-P-0004032-T01-IM5-20, EVT-P-0005121-T02-IM5-21, EVT-P-0009097-T01-IM5-2, EVT-P-0012575-T01-IM5-3, EVT-P-0013777-T01-IM5-4, ... (35 total)
1587	1687	100	14	25	0	21	0.0	2.028571784440879e-07	6.692809619443533	EVT-P-0005121-T02-IM5-21, EVT-P-0012575-T01-IM5-3, EVT-P-0014381-T01-IM6-6, EVT-P-0015540-T01-IM6-7, EVT-P-0016557-T01-IM6-8, ... (14 total)
1593	1643	50	13	26	0	21	0.0	1.054857327909257e-06	5.976806275808734	EVT-P-0005121-T02-IM5-21, EVT-P-0012575-T01-IM5-3, EVT-P-0014381-T01-IM6-6, EVT-P-0015540-T01-IM6-7, EVT-P-0016557-T01-IM6-8, ... (13 total)
1618	1643	25	13	26	0	21	0.0	1.054857327909257e-06	5.976806275808734	EVT-P-0005121-T02-IM5-21, EVT-P-0012575-T01-IM5-3, EVT-P-0014381-T01-IM6-6, EVT-P-0015540-T01-IM6-7, EVT-P-0016557-T01-IM6-8, ... (13 total)
1630	1730	100	15	24	1	20	0.014285714285714285	4.746857975591655e-06	5.32359376203339	EVT-P-0005121-T02-IM5-21, EVT-P-0012575-T01-IM5-3, EVT-P-0014381-T01-IM6-6, EVT-P-0015540-T01-IM6-7, EVT-P-0016557-T01-IM6-8, ... (15 total)
1637	1687	50	14	25	0	21	0.0	2.028571784440879e-07	6.692809619443533	EVT-P-0005121-T02-IM5-21, EVT-P-0012575-T01-IM5-3, EVT-P-0014381-T01-IM6-6, EVT-P-0015540-T01-IM6-7, EVT-P-0016557-T01-IM6-8, ... (14 total)
1643	1668	25	13	26	0	21	0.0	1.054857327909257e-06	5.976806275808734	EVT-P-0005121-T02-IM5-21, EVT-P-0012575-T01-IM5-3, EVT-P-0014381-T01-IM6-6, EVT-P-0015540-T01-IM6-7, EVT-P-0016557-T01-IM6-8, ... (13 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
1643	1693	50	14	25	0	21	0.0	2.028571784440879e-07	6.692809619443533	EVT-P-0005121-T02-IM5-21, EVT-P-0012575-T01-IM5-3, EVT-P-0014381-T01-IM6-6, EVT-P-0015540-T01-IM6-7, EVT-P-0016557-T01-IM6-8, ... (14 total)
1643	1743	100	15	24	1	20	0.014285714285714285	4.746857975591655e-06	5.32359376203339	EVT-P-0005121-T02-IM5-21, EVT-P-0012575-T01-IM5-3, EVT-P-0014381-T01-IM6-6, EVT-P-0015540-T01-IM6-7, EVT-P-0016557-T01-IM6-8, ... (15 total)
1643	1843	200	35	4	21	0	None	0.03720319509793194	1.4294197602844056	EVT-P-0004032-T01-IM5-20, EVT-P-0005121-T02-IM5-21, EVT-P-0009097-T01-IM5-2, EVT-P-0012575-T01-IM5-3, EVT-P-0013777-T01-IM5-4, ... (35 total)
1649	1749	100	21	18	20	1	340.0	6.077697283028405e-09	8.216260934987297	EVT-P-0004032-T01-IM5-20, EVT-P-0009097-T01-IM5-2, EVT-P-0013777-T01-IM5-4, EVT-P-0014191-T01-IM6-5, EVT-P-0019127-T01-IM6-11, ... (21 total)
1654	1754	100	22	17	21	0	None	3.527950929462398e-10	9.452477464133164	EVT-P-0004032-T01-IM5-20, EVT-P-0009097-T01-IM5-2, EVT-P-0013777-T01-IM5-4, EVT-P-0014191-T01-IM6-5, EVT-P-0019127-T01-IM6-11, ... (22 total)
1687	1787	100	22	17	21	0	None	3.527950929462398e-10	9.452477464133164	EVT-P-0004032-T01-IM5-20, EVT-P-0009097-T01-IM5-2, EVT-P-0013777-T01-IM5-4, EVT-P-0014191-T01-IM6-5, EVT-P-0019127-T01-IM6-11, ... (22 total)
1687	1887	200	22	17	21	0	None	3.527950929462398e-10	9.452477464133164	EVT-P-0004032-T01-IM5-20, EVT-P-0009097-T01-IM5-2, EVT-P-0013777-T01-IM5-4, EVT-P-0014191-T01-IM6-5, EVT-P-0019127-T01-IM6-11, ... (22 total)
1699	1749	50	20	19	20	1	None	3.0468667118084345e-10	9.516146544002542	EVT-P-0004032-T01-IM5-20, EVT-P-0009097-T01-IM5-2, EVT-P-0013777-T01-IM5-4, EVT-P-0014191-T01-IM6-5, EVT-P-0019127-T01-IM6-11, ... (20 total)
1704	1754	50	21	18	21	0	None	1.6036140588465447e-11	10.79490014495537	EVT-P-0004032-T01-IM5-20, EVT-P-0009097-T01-IM5-2, EVT-P-0013777-T01-IM5-4, EVT-P-0014191-T01-IM6-5, EVT-P-0019127-T01-IM6-11, ... (21 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
1724	1749	25	20	19	20	1	None	3.0468667118084345e-10	9.516146544002542	EVT-P-0004032-T01-IM5-20, EVT-P-0009097-T01-IM5-2, EVT-P-0013777-T01-IM5-4, EVT-P-0014191-T01-IM6-5, EVT-P-0019127-T01-IM6-11, ... (20 total)
1729	1754	25	21	18	21	0	None	1.6036140588465447e-11	10.79490014495537	EVT-P-0004032-T01-IM5-20, EVT-P-0009097-T01-IM5-2, EVT-P-0013777-T01-IM5-4, EVT-P-0014191-T01-IM6-5, EVT-P-0019127-T01-IM6-11, ... (21 total)
1730	1755	25	21	18	21	0	None	1.6036140588465447e-11	10.79490014495537	EVT-P-0004032-T01-IM5-20, EVT-P-0009097-T01-IM5-2, EVT-P-0013777-T01-IM5-4, EVT-P-0014191-T01-IM6-5, EVT-P-0019127-T01-IM6-11, ... (21 total)
1730	1780	50	21	18	21	0	None	1.6036140588465447e-11	10.79490014495537	EVT-P-0004032-T01-IM5-20, EVT-P-0009097-T01-IM5-2, EVT-P-0013777-T01-IM5-4, EVT-P-0014191-T01-IM6-5, EVT-P-0019127-T01-IM6-11, ... (21 total)
1730	1830	100	21	18	21	0	None	1.6036140588465447e-11	10.79490014495537	EVT-P-0004032-T01-IM5-20, EVT-P-0009097-T01-IM5-2, EVT-P-0013777-T01-IM5-4, EVT-P-0014191-T01-IM6-5, EVT-P-0019127-T01-IM6-11, ... (21 total)
1730	1930	200	21	18	21	0	None	1.6036140588465447e-11	10.79490014495537	EVT-P-0004032-T01-IM5-20, EVT-P-0009097-T01-IM5-2, EVT-P-0013777-T01-IM5-4, EVT-P-0014191-T01-IM6-5, EVT-P-0019127-T01-IM6-11, ... (21 total)
1749	1774	25	20	19	20	1	None	3.0468667118084345e-10	9.516146544002542	EVT-P-0004032-T01-IM5-20, EVT-P-0009097-T01-IM5-2, EVT-P-0013777-T01-IM5-4, EVT-P-0014191-T01-IM6-5, EVT-P-0019127-T01-IM6-11, ... (20 total)
1749	1799	50	20	19	20	1	None	3.0468667118084345e-10	9.516146544002542	EVT-P-0004032-T01-IM5-20, EVT-P-0009097-T01-IM5-2, EVT-P-0013777-T01-IM5-4, EVT-P-0014191-T01-IM6-5, EVT-P-0019127-T01-IM6-11, ... (20 total)
1749	1849	100	20	19	20	1	None	3.0468667118084345e-10	9.516146544002542	EVT-P-0004032-T01-IM5-20, EVT-P-0009097-T01-IM5-2, EVT-P-0013777-T01-IM5-4, EVT-P-0014191-T01-IM6-5, EVT-P-0019127-T01-IM6-11, ... (20 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
1749	1949	200	20	19	20	1	None	3.0468667118084345e-10	9.516146544002542	EVT-P-0004032-T01-IM5-20, EVT-P-0009097-T01-IM5-2, EVT-P-0013777-T01-IM5-4, EVT-P-0014191-T01-IM6-5, EVT-P-0019127-T01-IM6-11, ... (20 total)

top_windows

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
1729	1754	25	21	18	21	0	None	1.6036140588465447e-11	10.79490014495537	EVT-P-0004032-T01-IM5-20, EVT-P-0009097-T01-IM5-2, EVT-P-0013777-T01-IM5-4, EVT-P-0014191-T01-IM6-5, EVT-P-0019127-T01-IM6-11, ... (21 total)
1724	1749	25	20	19	20	1	None	3.0468667118084345e-10	9.516146544002542	EVT-P-0004032-T01-IM5-20, EVT-P-0009097-T01-IM5-2, EVT-P-0013777-T01-IM5-4, EVT-P-0014191-T01-IM6-5, EVT-P-0019127-T01-IM6-11, ... (20 total)
1749	1774	25	20	19	20	1	None	3.0468667118084345e-10	9.516146544002542	EVT-P-0004032-T01-IM5-20, EVT-P-0009097-T01-IM5-2, EVT-P-0013777-T01-IM5-4, EVT-P-0014191-T01-IM6-5, EVT-P-0019127-T01-IM6-11, ... (20 total)
1654	1754	100	22	17	21	0	None	3.527950929462398e-10	9.452477464133164	EVT-P-0004032-T01-IM5-20, EVT-P-0009097-T01-IM5-2, EVT-P-0013777-T01-IM5-4, EVT-P-0014191-T01-IM6-5, EVT-P-0019127-T01-IM6-11, ... (22 total)
1649	1749	100	21	18	20	1	340.0	6.077697283028405e-09	8.216260934987297	EVT-P-0004032-T01-IM5-20, EVT-P-0009097-T01-IM5-2, EVT-P-0013777-T01-IM5-4, EVT-P-0014191-T01-IM6-5, EVT-P-0019127-T01-IM6-11, ... (21 total)
1637	1687	50	14	25	0	21	0.0	2.028571784440879e-07	6.692809619443533	EVT-P-0005121-T02-IM5-21, EVT-P-0012575-T01-IM5-3, EVT-P-0014381-T01-IM6-6, EVT-P-0015540-T01-IM6-7, EVT-P-0016557-T01-IM6-8, ... (14 total)
1443	1643	200	14	25	0	21	0.0	2.028571784440879e-07	6.692809619443533	EVT-P-0005121-T02-IM5-21, EVT-P-0012575-T01-IM5-3, EVT-P-0014381-T01-IM6-6, EVT-P-0015540-T01-IM6-7, EVT-P-0016557-T01-IM6-8, ... (14 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
1618	1643	25	13	26	0	21	0.0	1.054857327909257e-06	5.976806275808734	EVT-P-0005121-T02-IM5-21, EVT-P-0012575-T01-IM5-3, EVT-P-0014381-T01-IM6-6, EVT-P-0015540-T01-IM6-7, EVT-P-0016557-T01-IM6-8, ... (13 total)
1630	1730	100	15	24	1	20	0.014285714285714285	4.746857975591655e-06	5.32359376203339	EVT-P-0005121-T02-IM5-21, EVT-P-0012575-T01-IM5-3, EVT-P-0014381-T01-IM6-6, EVT-P-0015540-T01-IM6-7, EVT-P-0016557-T01-IM6-8, ... (15 total)
1554	1754	200	35	4	21	0	None	0.03720319509793194	1.4294197602844056	EVT-P-0004032-T01-IM5-20, EVT-P-0005121-T02-IM5-21, EVT-P-0009097-T01-IM5-2, EVT-P-0012575-T01-IM5-3, EVT-P-0013777-T01-IM5-4, ... (35 total)
1249	1449	200	4	35	0	21	0.0	0.03720319509793195	1.4294197602844054	EVT-P-0021127-T01-IM6-13, EVT-P-0034117-T01-IM6-1, EVT-P-0035844-T03-IM7-23, EVT-P-0064872-T01-IM7-39
1549	1749	200	34	5	20	1	5.714285714285714	0.16183389867600395	0.7909305033297682	EVT-P-0004032-T01-IM5-20, EVT-P-0005121-T02-IM5-21, EVT-P-0009097-T01-IM5-2, EVT-P-0012575-T01-IM5-3, EVT-P-0013777-T01-IM5-4, ... (34 total)

Per-event results (results.tsv)

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0034117-T01-IM6-1	P-0034117-T01-IM6	P-0034117	Breast Cancer	BRCNOS	KIAA1549-BRAF	BRAF	in-frame	five_prime	138556165	12	1449	True	1448	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.621257485029402	True			UPF0606 protein KIAA1549 alpha/beta domain 2	KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4346-57:KIAA1549_c.1178-855:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0009097-T01-IM5-2	P-0009097-T01-IM5	P-0009097	Glioma	PAST	KIAA1549-BRAF	BRAF	in-frame	five_prime	138542048	16	1749	True	1749	True	exact match	retained	1.0	False	UPF0606 protein KIAA1549 alpha/beta domain 2			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused in frame with BRAF exons 9-18, which include the BRAF kinase domain): c.5247+3837:KIAA1549_c.1141-1433:BRAFdup | Protein fusion: in frame (KIAA1549-BRAF)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0012575-T01-IM5-3	P-0012575-T01-IM5	P-0012575	Glioma	PAST	KIAA1549-BRAF	BRAF	in-frame	five_prime	138551928	15	1643	True	1643	True	exact match	disrupted	0.911676646706 5869	True			UPF0606 protein KIAA1549 alpha/beta domain 2	KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 with BRAF exons 9-18): c.4929+793:KIAA1549_c.1141-2911:BRAFdup | Protein fusion: in frame (KIAA1549-BRAF)	NA
EVT-P-0013777-T01-IM5-4	P-0013777-T01-IM5	P-0013777	Glioma	PAST	KIAA1549-BRAF	BRAF	in-frame	five_prime	138539749	16	1749	True	1749	True	exact match	retained	1.0	False	UPF0606 protein KIAA1549 alpha/beta domain 2			KIAA1549(NM_001164665) - BRAF (NM_004333) rearrangement: c.5248-2734:KIAA1549_c.1315-181:BRAFdup | Protein fusion: in frame (KIAA1549-BRAF)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0014191-T01-IM6-5	P-0014191-T01-IM6	P-0014191	Glioma	PAST	KIAA1549-BRAF	BRAF	in-frame	five_prime	138537335	16	1749	True	1749	True	exact match	retained	1.0	False	UPF0606 protein KIAA1549 alpha/beta domain 2			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion : c.5248-320: KIAA1549_c.1140+1429:BRAFdup | Protein fusion: in frame (KIAA1549-BRAF)	NA
EVT-P-0014381-T01-IM6-6	P-0014381-T01-IM6	P-0014381	Glioma	PAST	KIAA1549-BRAF	BRAF	in-frame	five_prime	138548284	15	1643	True	1643	True	exact match	disrupted	0.911676646706 5869	True			UPF0606 protein KIAA1549 alpha/beta domain 2	KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 fused to BRAF exon 9 -18): c.4930-2082:KIAA1549_c.1140+1220:BRAFdup | Protein fusion: in frame (KIAA1549-BRAF)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0015540-T01-IM6-7	P-0015540-T01-IM6	P-0015540	Glioma	GB	KIAA1549-BRAF	BRAF	in-frame	five_prime	138546271	15	1643	True	1643	True	exact match	disrupted	0.911676646706 5869	True			UPF0606 protein KIAA1549 alpha/beta domain 2	KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 fused in frame with BRAF exons 9-18, which include the BRAF kinase domain): c.4930-69:KIAA1549_c.1141-1735:BRAFdup | Protein fusion: in frame (KIAA1549-BRAF)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0016557-T01-IM6-8	P-0016557-T01-IM6	P-0016557	Glioma	PAST	KIAA1549-BRAF	BRAF	in-frame	five_prime	138549766	15	1643	True	1643	True	exact match	disrupted	0.911676646706 5869	True			UPF0606 protein KIAA1549 alpha/beta domain 2	KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 fused in-frame to BRAF exons 9-18) : c.4929+2955:KIAA1549_c.1141-1192:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0017693-T01-IM6-9	P-0017693-T01-IM6	P-0017693	Glioma	GB	KIAA1549-BRAF	BRAF	in-frame	five_prime	138546697	15	1643	True	1643	True	exact match	disrupted	0.911676646706 5869	True			UPF0606 protein KIAA1549 alpha/beta domain 2	KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (c.1141-2153): c.4930-495:KIAA1549_c.1141-2153:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0017693-T02-IM6-10	P-0017693-T02-IM6	P-0017693	Glioma	PAST	KIAA1549-BRAF	BRAF	in-frame	five_prime	138546697	15	1643	True	1643	True	exact match	disrupted	0.911676646706 5869	True			UPF0606 protein KIAA1549 alpha/beta domain 2	KIAA1549 (NM_001164665) - BRAF (NM_004333) Fusion: c.4930-495:KIAA1549_c.1141-2153:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0019127-T01-IM6-11	P-0019127-T01-IM6	P-0019127	Glioma	PAST	KIAA1549-BRAF	BRAF	in-frame	five_prime	138543775	16	1749	True	1749	True	exact match	retained	1.0	False	UPF0606 protein KIAA1549 alpha/beta domain 2			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused to BRAF exons 9-18): c.5247+2110:KIAA1549_c.1141-3273:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0020125-T01-IM6-12	P-0020125-T01-IM6	P-0020125	Glioma	PAST	KIAA1549-BRAF	BRAF	unknown	five_prime	138537003	17	1754	False	1753	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	UPF0606 protein KIAA1549 alpha/beta domain 2			KIAA1549 (NM_001164665) - BRAF (NM_00433) fusion (KIAA1549 exons 1-17 fused to BRAF exons 9-18): c.5260:KIAA1549_c.1140+1933:BRAFdup | Protein Fusion: mid-exon {KIAA1549:BRAF}	NA
EVT-P-0021127-T01-IM6-13	P-0021127-T01-IM6	P-0021127	Breast Cancer	ILC	KIAA1549-BRAF	BRAF	in-frame	five_prime	138575092	10	1344	True	1344	True	exact match	disrupted	0.4640718562874251	True			UPF0606 protein KIAA1549 alpha/beta domain 2	KIAA1549 (NM_001164665) - BRAF (NM_00433) fusion (KIAA1549 exons 1-10 fused with BRAF exons 9-18) : c.4032+3996:KIAA1549_c.1140+2367:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0022404-T01-IM6-14	P-002240404-T01-IM6	P-00224004	Glioma	PAST	KIAA1549-BRAF	BRAF	in-frame	five_prime	138539285	16	1749	True	1749	True	exact match	retained	1.0	False	UPF0606 protein KIAA1549 alpha/beta domain 2			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused with BRAF exons 11-18) : c.5248-2270:KIAA1549_c.1314+644:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0027592-T01-IM6-15	P-002759292-T01-IM6	P-00275992	Glioma	PAST	KIAA1549-BRAF	BRAF	in-frame	five_prime	138545116	16	1749	True	1749	True	exact match	retained	1.0	False	UPF0606 protein KIAA1549 alpha/beta domain 2			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused with BRAF exons 9-18) : c.5247+769:KIAA1549_c.1140+1765:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0028611-T01-IM6-16	P-0028611-T01-IM6	P-0028611	Glioma	PAST	KIAA1549-BRAF	BRAF	in-frame	five_prime	138541164	16	1749	True	1749	True	exact match	retained	1.0	False	UPF0606 protein KIAA1549 alpha/beta domain 2			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused to BRAF exons 11-18): c.5248-4149:KIAA1549_c.1314+35:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0028611-T02-IM6-17	P-0028611-T02-IM6	P-0028611	Glioma	PAST	KIAA1549-BRAF	BRAF	in-frame	five_prime	138541164	16	1749	True	1749	True	exact match	retained	1.0	False	UPF0606 protein KIAA1549 alpha/beta domain 2			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused to BRAF exons 11-18): c.5248-4149:KIAA1549_c.1314+35:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0029361-T01-IM6-18	P-0029361-T01-IM6	P-0029361	Glioma	PAST	KIAA1549-BRAF	BRAF	in-frame	five_prime	138548869	15	1643	True	1643	True	exact match	disrupted	0.911676646706 5869	True			UPF0606 protein KIAA1549 alpha/beta domain 2	KIAA1549 (NM_001164665) - BRAF (NM_004333) Fusion: c.4930-2667:KIAA1549_c.1141-2241:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0032982-T01-IM6-19	P-0032982-T01-IM6	P-0032982	Glioma	PAST	KIAA1549-BRAF	BRAF	in-frame	five_prime	138537119	16	1749	True	1749	True	exact match	retained	1.0	False	UPF0606 protein KIAA1549 alpha/beta domain 2			KIAA1549 (NM_001164665) - BRAF (NM_004333) Fusion (KIAA1549 exons 1-16 fused with BRAF exons 9-18) : c.5248-10:KIAA1549_c.1140+2602:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0004032-T01-IM5-20	P-0004032-T01-IM5	P-0004032	Glioma	AGNG	KIAA1549-BRAF	BRAF	in-frame	five_prime	138538964	16	1749	True	1749	True	exact match	retained	1.0	False	UPF0606 protein KIAA1549 alpha/beta domain 2			KIAA1549 (NM_001164665) - BRAF (NM_004333) Fusion (KIAA1549 exon 16 fused to BRAF exon 9) : c.5248-1949:KIAA1549_c.1141-508:BRAFdup | Protein fusion: in frame (KIAA1549-BRAF)	NA
EVT-P-0005121-T02-IM5-21	P-0005121-T02-IM5	P-0005121	Melanoma	ACRM	KIAA1549-BRAF	BRAF	in-frame	five_prime	138550267	15	1643	True	1643	True	exact match	disrupted	0.911676646706 5869	True			UPF0606 protein KIAA1549 alpha/beta domain 2	KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 with BRAF exons 9-18): c.4929+2454:KIAA1549_c.1140+2860:BRAFdup | Protein fusion: in frame (KIAA1549-BRAF)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0035153-T03-IM6-22	P-0035153-T03-IM6	P-0035153	Glioma	GB	KIAA1549-BRAF	BRAF	in-frame	five_prime	138548248	15	1643	True	1643	True	exact match	disrupted	0.911676646706 5869	True			UPF0606 protein KIAA1549 alpha/beta domain 2	KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4930-2046:KIAA1549_c.1140+3015:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0035844-T03-IM7-23	P-0035844-T03-IM7	P-0035844	Soft Tissue Sarcoma	ERMS	KIAA1549-BRAF	BRAF	in-frame	five_prime	138567532	10	1344	True	1344	True	exact match	disrupted	0.464071856287 42514	True			UPF0606 protein KIAA1549 alpha/beta domain 2	KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4033-1202:KIAA1549_c.1140+823:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0035844-T04-IM7-24	P-0035844-T04-IM7	P-0035844	Soft Tissue Sarcoma	ERMS	KIAA1549-BRAF	BRAF	in-frame	five_prime	138567532	10	1344	True	1344	True	exact match	disrupted	0.4640718562874251 4	True			UPF0606 protein KIAA1549 alpha/beta domain 2	KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4033-1202:KIAA1549_c.1140+823:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0040760-T01-IM6-25	P-0040760-T01-IM6	P-0040760	Glioma	PAST	KIAA1549-BRAF	BRAF	in-frame	five_prime	138550094	15	1643	True	1643	True	exact match	disrupted	0.9116766467065869	True			UPF0606 protein KIAA1549 alpha/beta domain 2	KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4929+2627:KIAA1549_c.1315-47:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0043372-T01-IM6-26	P-0043372-T01-IM6	P-0043372	Miscellaneous Neuroepithelial Tumor	EVN	KIAA1549-BRAF	BRAF	in-frame	five_prime	138550517	15	1643	True	1643	True	exact match	disrupted	0.911676646706 5869	True			UPF0606 protein KIAA1549 alpha/beta domain 2	KIAA1549 (NM_001164665) - BRAF (NM_00433) fusion: c.4929+2204:KIAA1549_c.1141-139:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0043874-T01-IM6-27	P-0043874-T01-IM6	P-0043874	Glioma	GNG	KIAA1549-BRAF	BRAF	in-frame	five_prime	138539922	16	1749	True	1749	True	exact match	retained	1.0	False	UPF0606 protein KIAA1549 alpha/beta domain 2			KIAA1549 (NM_001164665) - BRAF (NM_00433) fusion: c.5248-2907:KIAA1549_c.1141-2617:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0044270-T02-IM6-28	P-0044270-T02-IM6	P-0044270	Glioma	PAST	KIAA1549-BRAF	BRAF	in-frame	five_prime	138544325	16	1749	True	1749	True	exact match	retained	1.0	False	UPF0606 protein KIAA1549 alpha/beta domain 2			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5247+1560:KIAA1549_c.1140+2463:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0045683-T02-IM6-29	P-0045683-T02-IM6	P-0045683	Glioma	PMA	KIAA1549-BRAF	BRAF	unknown	five_prime	138545943	16	1730	False	1729	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	UPF0606 protein KIAA1549 alpha/beta domain 2			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5189:KIAA1549_c.1140+1258:BRAFdup | Protein Fusion: mid-exon {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0049925-T01-IM6-30	P-0049925-T01-IM6	P-0049925	Glioma	PAST	KIAA1549-BRAF	BRAF	in-frame	five_prime	138545584	16	1749	True	1749	True	exact match	retained	1.0	False	UPF0606 protein KIAA1549 alpha/beta domain 2			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5247+301:KIAA1549_c.1141-2113:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0052411-T03-IM6-31	P-0052411-T03-IM6	P-0052411	Miscellaneous Neuroepithelial Tumor	MNET	KIAA1549-BRAF	BRAF	in-frame	five_prime	138548498	15	1643	True	1643	True	exact match	disrupted	0.911676646706 5869	True			UPF0606 protein KIAA1549 alpha/beta domain 2	KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4930-2296:KIAA1549_c.1140+578:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0052411-T05-IM7-32	P-0052411-T05-IM7	P-0052411	Miscellaneous Neuroepithelial Tumor	MNET	KIAA1549-BRAF	BRAF	in-frame	five_prime	138548498	15	1643	True	1643	True	exact match	disrupted	0.911676646706 5869	True			UPF0606 protein KIAA1549 alpha/beta domain 2	KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4930-2296:KIAA1549_c.1140+578:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0055099-T03-IM6-33	P-0055099-T03-IM6	P-0055099	Miscellaneous Neuroepithelial Tumor	MNET	KIAA1549-BRAF	BRAF	in-frame	five_prime	138551364	15	1643	True	1643	True	exact match	disrupted	0.911676646706 5869	True			UPF0606 protein KIAA1549 alpha/beta domain 2	KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4929+1357:KIAA1549_c.1141-3279:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0056952-T01-IM6-34	P-0056952-T01-IM6	P-0056952	Nerve Sheath Tumor	MPNST	KIAA1549-BRAF	BRAF	in-frame	five_prime	138537156	16	1749	True	1749	True	exact match	retained	1.0	False	UPF0606 protein KIAA1549 alpha/beta domain 2			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5248-141:KIAA1549_c.1140+287:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0058584-T01-IM7-35	P-0058584-T01-IM7	P-0058584	Nerve Sheath Tumor	NST	KIAA1549-BRAF	BRAF	in-frame	five_prime	138544222	16	1749	True	1749	True	exact match	retained	1.0	False	UPF0606 protein KIAA1549 alpha/beta domain 2			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5247+1663:KIAA1549_c.1140+2615:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0062070-T01-IM7-36	P-0062070-T01-IM7	P-0062070	Glioma	GB	KIAA1549-BRAF	BRAF	in-frame	five_prime	138538079	16	1749	True	1749	True	exact match	retained	1.0	False	UPF0606 protein KIAA1549 alpha/beta domain 2			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5248-1064:KIAA1549_c.1141-3059:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0062963-T01-IM7-37	P-0062963-T01-IM7	P-0062963	Glioma	PAST	KIAA1549-BRAF	BRAF	in-frame	five_prime	138546229	15	1643	True	1643	True	exact match	disrupted	0.911676646706 5869	True			UPF0606 protein KIAA1549 alpha/beta domain 2	KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4930-27:KIAA1549_c.1314+623:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0064310-T04-IM7-38	P-0064310-T04-IM7	P-0064310	Glioma	PAST	KIAA1549-BRAF	BRAF	in-frame	five_prime	138542325	16	1749	True	1749	True	exact match	retained	1.0	False	UPF0606 protein KIAA1549 alpha/beta domain 2			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5247+3560:KIAA1549_c.1140+662:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0064872-T01-IM7-39	P-0064872-T01-IM7	P-0064872	Glioma	GB	KIAA1549-BRAF	BRAF	in-frame	five_prime	138574741	10	1344	True	1344	True	exact match	disrupted	0.4640718562874251 4	True			UPF0606 protein KIAA1549 alpha/beta domain 2	KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4032+4347:KIAA1549_c.1141-1893:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA

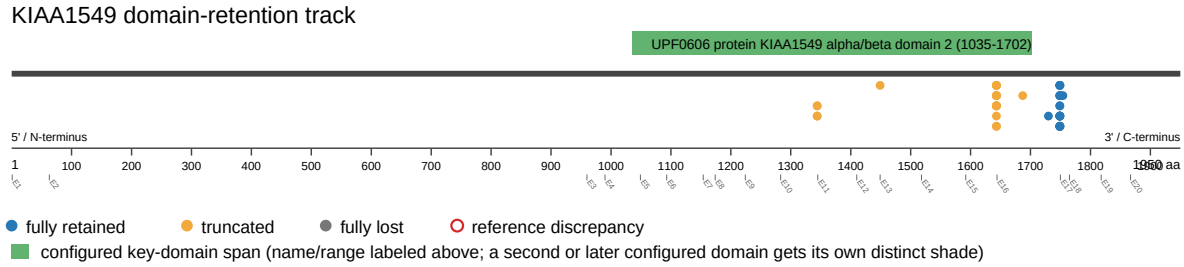
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0066022-T01-IM7-40	P-0066022-T01-IM7	P-0066022	Glioma	PAST	KIAA1549-BRAF	BRAF	in-frame	five_prime	138541089	16	1749	True	1749	True	exact match	retained	1.0	False	UPF0606 protein KIAA1549 alpha/beta domain 2			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5248-4074:KIAA1549_c.1141-651:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0066030-T01-IM7-41	P-0066030-T01-IM7	P-0066030	Glioma	PAST	KIAA1549-BRAF	BRAF	in-frame	five_prime	138538187	16	1749	True	1749	True	exact match	retained	1.0	False	UPF0606 protein KIAA1549 alpha/beta domain 2			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5248-1172:KIAA1549_c.1314+36:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0069024-T04-IM7-42	P-0069024-T04-IM7	P-0069024	Glioma	PAST	KIAA1549-BRAF	BRAF	unknown	five_prime	138546073	16	1687	False	1686	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.977544910179 6407	True			UPF0606 protein KIAA1549 alpha/beta domain 2	KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5059:KIAA1549_c.1140+1991:BRAFdup | Protein Fusion: mid-exon {KIAA1549:BRAF}	NA
EVT-P-0035153-T01-IM6-43	P-0035153-T01-IM6	P-0035153	Glioma	GB	KIAA1549-BRAF	BRAF	in-frame	five_prime	138548248	15	1643	True	1643	True	exact match	disrupted	0.911676646706 5869	True			UPF0606 protein KIAA1549 alpha/beta domain 2	KIAA1549 (NM_001164665) - BRAF (NM_004333) Fusion (KIAA1549 exon15 fused with BRAF exon9) : c.4930-2046:KIAA1549_c.1140+3015:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA

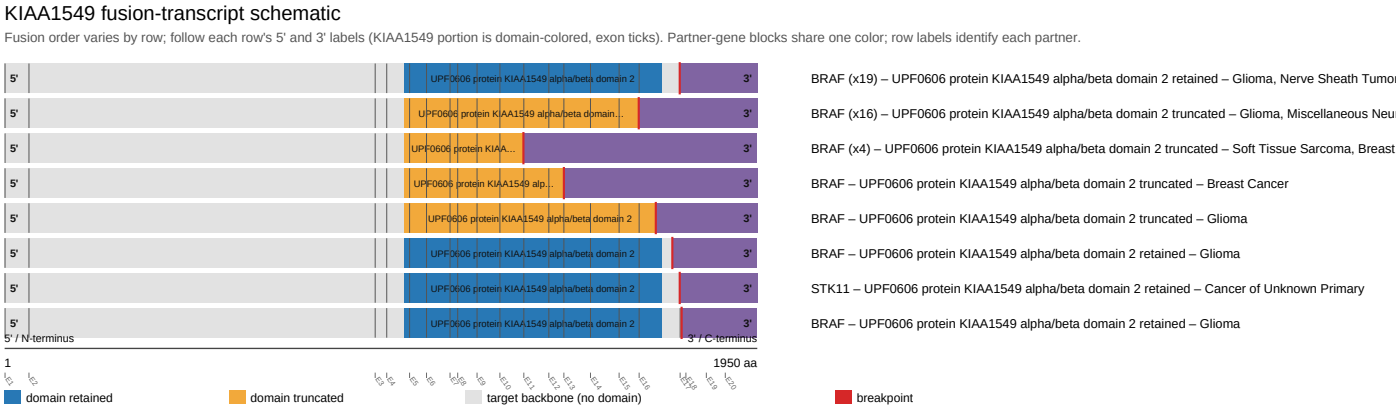
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0038818-T01-IM6-44	P-0038818-T01-IM6	P-0038818	Cancer of Unknown Primary	CUP	KIAA1549-STK11	STK11	out-of-frame	five_prime	138539377	16	1749	True	1749	True	exact match	retained	1.0	False	UPF0606 protein KIAA1549 alpha/beta domain 2			KIAA1549 (NM_001164665) - STK11 (NM_000455) Rearrangement : t(7,19)(q34;p13.3)(chr7:g.138539377::chr19:g.1221181) | Protein Fusion: out of frame {KIAA1549:STK11}	NA

Figures

domain retention outliers



fusion schematic



Partner-block colors are decorative; row labels identify partners.